

(1a) A scientific approach...

- a. compares observations with predictions
- b. only considers cultural facts
- c. requires gathering lots of information

(1b) Consider the hypothesis "Mary's machine learning algorithm is better than John's"...

- a. Running the experiment more than once will ensure generalisability
- b. 'Better' needs to be defined more carefully
- c. Testing this hypothesis means running both algorithms with different data

(1c) Considering ethics in human experimentation...

- a. means allowing participants to take as long as they like on an experimental task
- b. ensures that any risk to the participants' well-being is reduced as much as possible
- c. is important to ensure that your paper will be published

(1d) A hypothesis...

- a. even if supported by evidence now, may be rejected later
- b. is always rephrased as a research question
- c. should always be phrased so that it is impossible to prove wrong

(1e) Describe 3 principles of GDPR and give an example of each principle in relation to the University's storage of data about students

(1f) Discuss 3 characteristics of a good research paper

---

(2) You have developed a new algorithm, called MegaMerge (MM), for joining multiple lists of numbers together then sorting them into ascending order. You believe it does this faster than your existing algorithm called KiloMerge.

(2a) State this belief as a testable hypothesis

(2b) Explain how it meets the essential criteria of a good hypothesis

(2c) Describe 2 ways in which the generalizability of this experiment might be increased

(2d) Identify 1 quantitative dependent variable for this experiment

---

(3) You have been given a set of 100 newly designed app icon designs that are not yet associated with any existing app. You have created an app to help people remember other people's names.

Design an experiment to answer the following research question:

*"Which icon should you use as the icon of your new app"*